Heaven’s Light is Our Guide

Dept. of Electrical & Computer Engineering (ECE),
Rajshahi University of Engineering & Technology

Course Code: ECE 2215
Course Title: Data Base Systems
Topic: Introduction to Database and DBMS

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Database

Database: An organized collection of data stored and accessed electronically.

Mainly two types of database –

1. Structured database
   - in table format
2. Unstructured database
   - document or key-value based

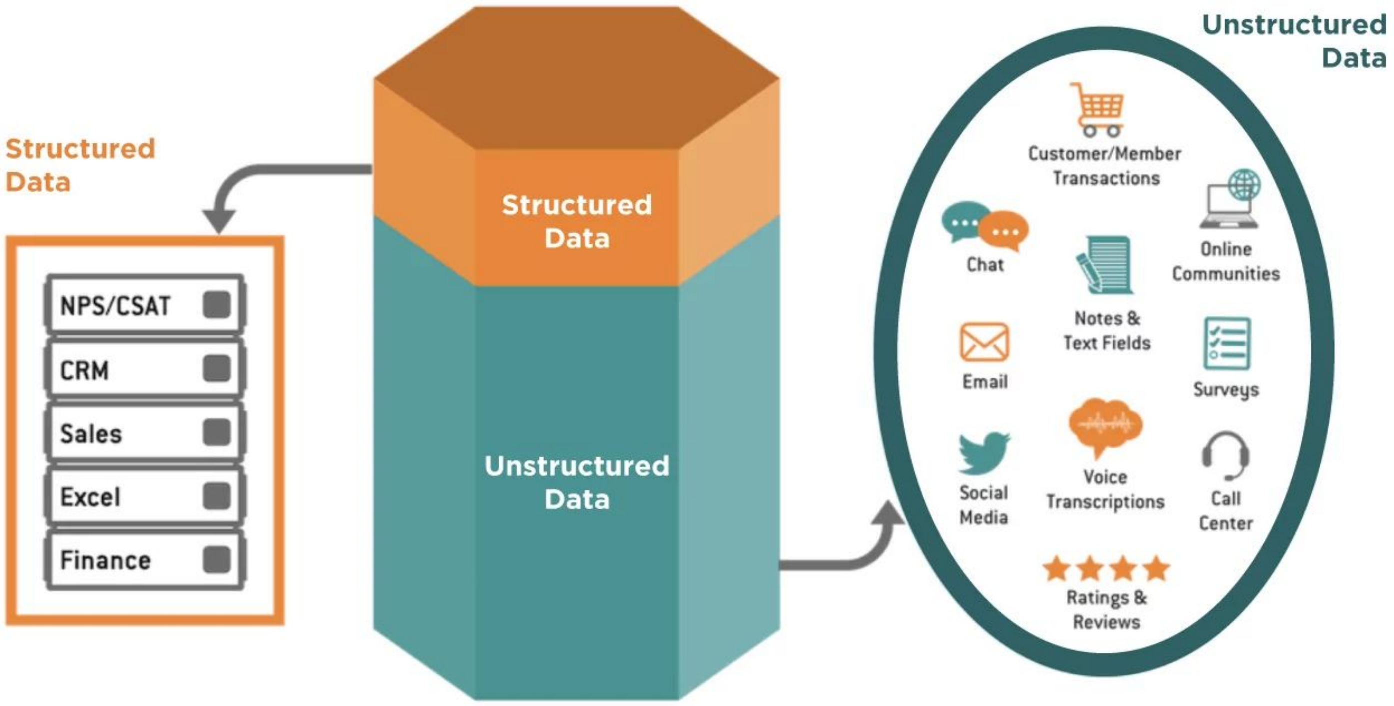


Fig. 1 Structured and Unstructured data

Data and Information

Data	Information
Data is unorganized and unrefined facts	Information comprises processed, organized data presented in a meaningful context
Data is an individual unit that contains raw materials which do not carry any specific meaning.	Information is a group of data that collectively carries a logical meaning.
Data doesn’t depend on information.	Information depends on data.
Raw data alone is insufficient for decision making	Information is sufficient for decision making
An example of data is a student’s test score	The average score of a class is the information derived from the given data.

Database Management System

- A database-management system (DBMS) is a set of programs to access the collection of interrelated data.

- DBMS contains information about a particular enterprise
  - Collection of interrelated data
  - Set of programs to access the data
  - An environment that is both convenient and efficient to use

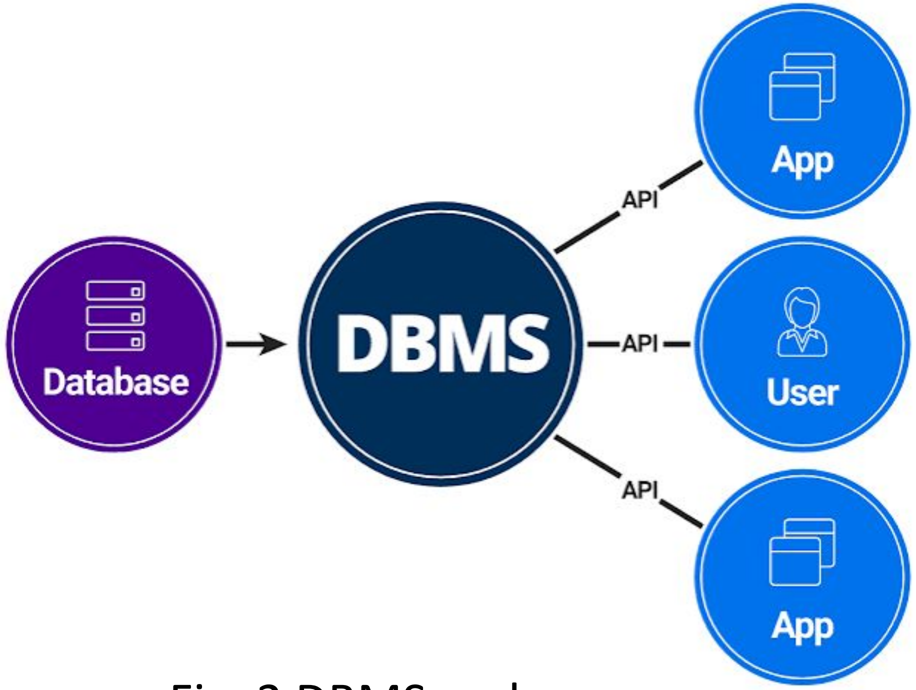


Fig. 2 DBMS and users

Database Application Examples

- Enterprise Information
  - Sales: customers, products, purchases
  - Accounting: payments, receipts, assets
  - Human Resources: Information about employees, salaries, payroll taxes.

- Manufacturing: management of production, inventory, orders, supply chain.

- Banking and finance
  - customer information, accounts, loans, and banking transactions.
  - Credit card transactions
  - Finance: sales and purchases of financial instruments (e.g., stocks and bonds; storing real-time market data

- Universities: registration, grades

Purpose of Database Systems

One way to keep the information on a computer is to store it in operating-system files, like - .txt, .docx, .xlsx etc.

Keeping organizational information in a file-processing system has a number of

major disadvantages:

- **Data redundancy:** data is stored in multiple file formats resulting in duplication of information in different files.
- **Data inconsistency:** The various copies of the same data may no longer agree.
- **Difficulty in accessing data:** Need to write a new program to carry out each new task
- **Data isolation:** Multiple files and formats.
- **Integrity problems:** When new constraints are added, it is difficult to change the programs to enforce them.

Purpose of Database Systems (Cont.)

- **Atomicity of updates**
  - Failures may leave the database in an inconsistent state with partial updates carried out
  - Example: Transfer of funds from one account to another should either be completed or not happen at all
- **Concurrent access by multiple users**
  - Concurrent access needed for performance
  - Uncontrolled concurrent accesses can lead to inconsistencies
    - Ex: Two people reading a balance (say 100) and updating it by withdrawing money (say 50 each) at the same time
- **Security problems**
  - Hard to provide user access to some, but not all, data.

Database systems offer solutions to all the above problems.

Components of DBMS

- **Data:**
  - User data
  - Metadata
- **Software**
  - Actual DBMS software like MySQL, Oracle, PostgreSQL.
- **Hardware**
  - Physical devices like servers, disks, input-output devices (keyboard, monitor, printer).
- **Procedure**
  - Instructions and rules for using DBMS effectively
- **Database Access Language**
  - Used to interact with the database (create, read, update, delete data).
  - Examples: SQL, MyAccess, Oracle PL/SQL.

Components of DBMS

- **Users:**
  - **Naïve Users:** The unsophisticated users who interact with the system by using predefined user interfaces, such as web or mobile applications.
  - **Application programmers** are computer professionals who write application programs. Application programmers can choose from many tools to develop user interfaces.
  - **Sophisticated users** interact with the system without writing programs. Instead, they form their requests either using a database query language or by using tools such as data analysis software.

Components of DBMS

- **Database Administrator:**

  A person who has central control over the system is called a **database administrator (DBA).** Functions of a DBA include:

  - Schema definition
  - Storage structure and access-method definition
  - Schema and physical-organization modification
  - Granting of authorization for data access
  - Routine maintenance
  - Periodically backing up the database
  - Ensuring that enough free disk space is available for normal operations, and upgrading disk space as required
  - Monitoring jobs running on the database

Data Models

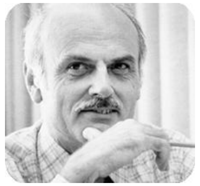
- Data models are a collection of tools for describing data, data relationships, data semantics and data constraints
    - Relational model
    - Entity-Relationship data model (mainly for database design)
    - Object-based data models (Object-oriented and Object-relational)
    - Semi-structured data model (XML)
    - Other older models:
        - Network model
        - Hierarchical model

Relational Models

- All the data is stored in various tables.
- Example of tabular data in the relational model

Columns

Rows



Ted Codd
Turing Award 1981

Levels of Abstraction

- A major purpose of a database system is to provide users with an abstract view of the data.

Data abstraction: Since many database-system users are not computer-trained, developers hide the complexity from users to simplify users' interactions with the system.

There are different levels of abstractions –

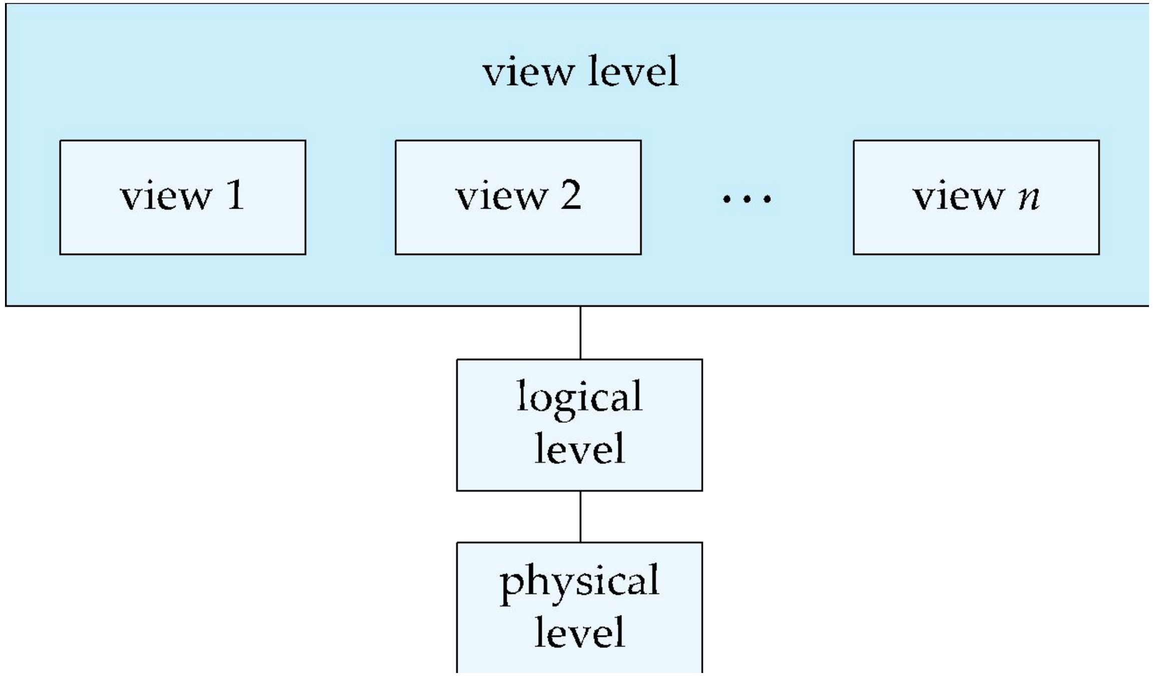
- **Physical level**: describes how a record (e.g., instructor) is stored.
- **Logical level**: describes data stored in database, and the relationships among the data.

```
type instructor = record
    ID : string;
    name : string;
    dept_name : string;
    salary : integer;
            end;
```

- **View level**: application programs hide details of data types. Views can also hide information (such as an employee's salary) for security purposes.

View of Data

An architecture for a database system



Instances and Schemas

- Similar to types and variables in programming languages
- Logical Schema – the overall logical structure of the database
  - Example: The database consists of information about a set of customers and accounts in a bank and the relationship between them
  - Analogous to type information of a variable in a program
- Physical schema – the overall physical structure of the database
- Instance – the actual content of the database at a particular point in time
  - Analogous to the value of a variable

Database Languages - DDL

- **Database languages** are specialized sets of commands and instructions used to define, manipulate and control data within a database.
- Specification notation for defining the database schema
- Example:

```
create table instructor (
    ID         char(5),
    name       varchar(20),
    dept_name  varchar(20),
    salary     numeric(8,2))
```

- DDL compiler generates a set of table templates stored in a data dictionary.

Database Languages - DDL

- **Database languages** are specialized sets of commands and instructions used to define, manipulate and control data within a database.
- Specification notation for defining the database schema – CREATE, DROP ALTER, RENAME etc.
- Example:

```
create table instructor (
    ID          char(5),
    name        varchar(20),
    dept_name   varchar(20),
    salary      numeric(8,2))
```

- DDL compiler generates a set of table templates stored in a data dictionary.

Database Languages - DML

- Language for accessing and updating the data organized by the appropriate data model. DML, also known as query language
- There are basically two types of data-manipulation language
    - **Procedural DML** requires a user to specify what data are needed and how to get that data. Example - Query: "Open a cursor, fetch the first employee record, check if their salary is >50000, if so, print the name, fetch the next record, and repeat until done."
    - **Declarative DML** requires a user to specify what data is needed without specifying how to get that data. Example - Query: "Find the names of all employees who earn more than $\$50,000$."
- Declarative DMLs are usually easier to learn and use than are procedural DMLs.
- Declarative DMLs are also referred to as non-procedural DMLs

Database Languages - DQL

- The portion of a DML that involves information retrieval is called a query language.
- SQL query language is nonprocedural. A query takes as input several tables (possibly only one) and always returns a single table.
- Example to find all instructors in Comp. Sci. dept

```
select name
from instructor
where dept_name = 'Comp. Sci.'
```

Database Access from Application Program

- Non-procedural query languages such as SQL are not as powerful as a universal Turing machine.
- SQL does not support actions such as input from users, output to displays, or communication over the network.
- Such computations and actions must be written in a host language, such as C/C++, Java or Python, with embedded SQL queries that access the data in the database.
- Application programs are programs that are used to interact with the database in this fashion.

Database Design

The process of designing the general structure of the database:

- Logical Design – Deciding on the database schema. Database design requires that we find a "good" collection of relation schemas.
    - Business decision – What attributes should we record in the database?
    - Computer Science decision – What relation schemas should we have and how should the attributes be distributed among the various relation schemas?
- Physical Design – Deciding on the physical layout of the database

Database Engine

- A database system is partitioned into modules that deal with each of the responsibilities of the overall system.
- The functional components of a database system can be divided into
  - The storage manager,
  - The query processor component,
  - The transaction management component.

Storage Manager

- A program module that provides the interface between the low-level data stored in the database and the application programs and queries submitted to the system.
- The storage manager is responsible to the following tasks:
  - Interaction with the OS file manager
  - Efficient storing, retrieving and updating of data
- The storage manager components include:
  - Authorization and integrity manager
  - Transaction manager
  - File manager
  - Buffer manager

Query Processor

- The query processor components include:
  - DDL interpreter - interprets DDL statements and records the definitions in the data dictionary.
  - DML compiler - translates DML statements in a query language into an evaluation plan consisting of low-level instructions that the query evaluation engine understands.
  - The DML compiler performs query optimization; that is, it picks the lowest-cost evaluation plan from among the various alternatives.
  - Query evaluation engine - executes low-level instructions generated by the DML compiler.

Query Processor

1. Parsing and translation
2. Optimization
3. Evaluation

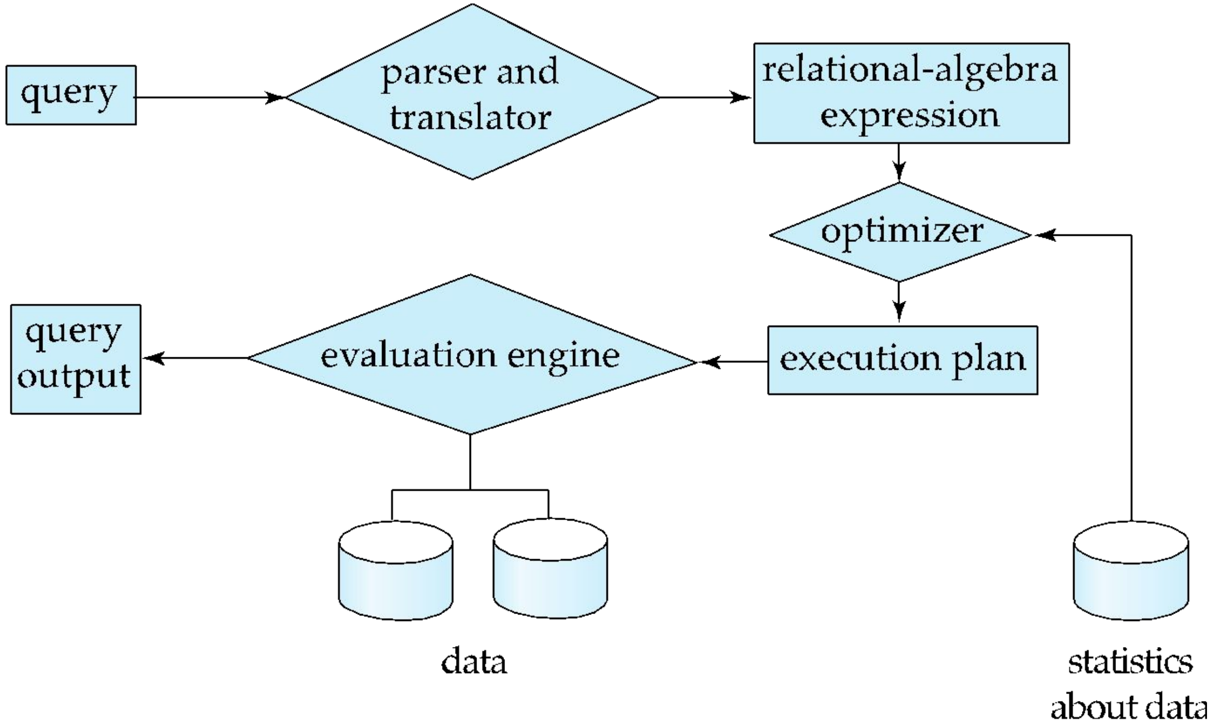


Fig. 3 Query processing

Transaction Management

- A transaction is a collection of operations that performs a single logical function in a database application
- Transaction-management component ensures that the database remains in a consistent (correct) state despite system failures (e.g., power failures and operating system crashes) and transaction failures.
- Concurrency-control manager controls the interaction among the concurrent transactions, to ensure the consistency of the database

Database Architecture

- Centralized databases
  - One to a few cores, shared memory
- Client-server,
  - One server machine executes work on behalf of multiple client machines.
- Parallel databases
  - Many core shared memory
  - Shared disk
  - Shared nothing
- Distributed databases
  - Geographical distribution
  - Schema/data heterogeneity

Database Architecture

Database Applications Architecture

- Database applications are usually partitioned into two or three parts
- Two-tier architecture -- the application resides at the client machine, where it invokes database system functionality at the server machine
- Three-tier architecture -- the client machine acts as a front end and does not contain any direct database calls.
    - The client end communicates with an application server, usually through a forms interface.
    - The application server in turn communicates with a database system to access data.

Database Applications Architecture

Database Applications Architecture

Thank You

If you have any queries, you can ask.